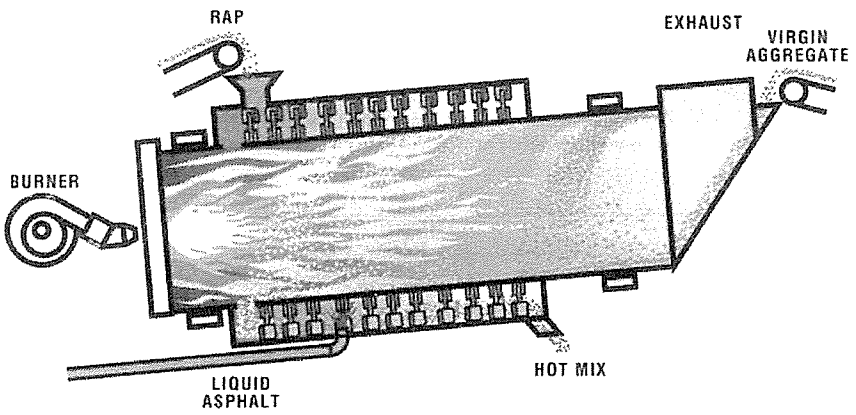


Figure 7.26 Double-Barrel Unitized Dryer/Mixer Flow

The driving force behind this design innovation in the 1980s was a desire to create a drum-mix facility that was more environmentally friendly by introducing the new liquid-asphalt binder outside of the main gas stream used to dry and heat the aggregate materials. This reduced the discharge of very small (1–5 micron) hydrocarbon particles in the stack gases, which sometimes created visible emissions or odors from the drum-mix facilities. These new drum-mixers eliminate this problem.

Figures 7.23 through 7.26 show the different styles of drum-mixers commonly found in the field at this time. It should be noted that many batch-plant operators are now converting their plants from batch plants to drum-mixers by replacing the tower with a mixing drum or continuous pugmill mixer or by bypassing the tower and using a small mixing drum mounted beside the tower, resulting in a hybrid system known as a combination plant. This allows operators to use the same dryer and baghouse to dry the material. As noted previously in this chapter, the cold-feed bins were probably already converted to belt feeders with variable-speed drives. By installing a continuous mixing device, a drum-mix liquid asphalt pumping and metering system, and drum-mix controls, the batch-plant operator now has all the advantages of drum-mix operation. Of primary interest to producers is the ability to run at the production rate possible with the dryer and not be limited by the capabilities of the screening unit, being able to run higher percentages of RAP, and reducing maintenance costs, thereby becoming more competitive in the marketplace. Having consistently sized aggregates available makes this conversion possible.

7.4.2 Aggregate Gradation Control in a Drum-Mixer

Controlling aggregate gradation with a drum-mixer is a much simpler process than with a batch plant. The individual cold-feed bins are calibrated for their speed/output function, and the final mix gradation is controlled

by proportioning these individual bins to the target mix formula. This, of course, depends on the availability of consistently sized individual aggregates from the aggregate supplier. If the individual aggregates are not consistent in gradation, then a consistent hot mix cannot be produced with a drum-mixer. Drum-mixers and the drum-mixing process depend on the availability of consistent aggregates to make high-quality mix.

The output from the cold-feed bins is really controlled in two ways. The primary adjustment is the variable-speed motor driving the belt feeder. In addition to this, the bins are fitted with adjustable gates. This procedure allows the producer to control the maximum and minimum flows from the bin. Some mixes require a large quantity of one material—in this case, the gate is opened wide. Some mixes require a small quantity of another material, and in this case the bin gate might be lowered. Variable-speed motors can be inconsistent in their flow at very low rates. They can also reach their maximum speed at the 95 percent, 96 percent, or 97 percent range. The most linear range is typically between 10 percent and 90 percent speed. Some are linear between 5 percent and 95 percent speed. It is best practice, therefore, to use the adjustable gate to ensure that the motor-drive combination is functioning in the 5–95 percent or 10–90 percent range at all times. Between these speed ranges, if output is charted to speed, the line usually has a consistent and predictable slope.

Figure 7.27 shows a typical calibration chart for a feed bin at different gate settings. Note that for very large quantities of material, the gate will

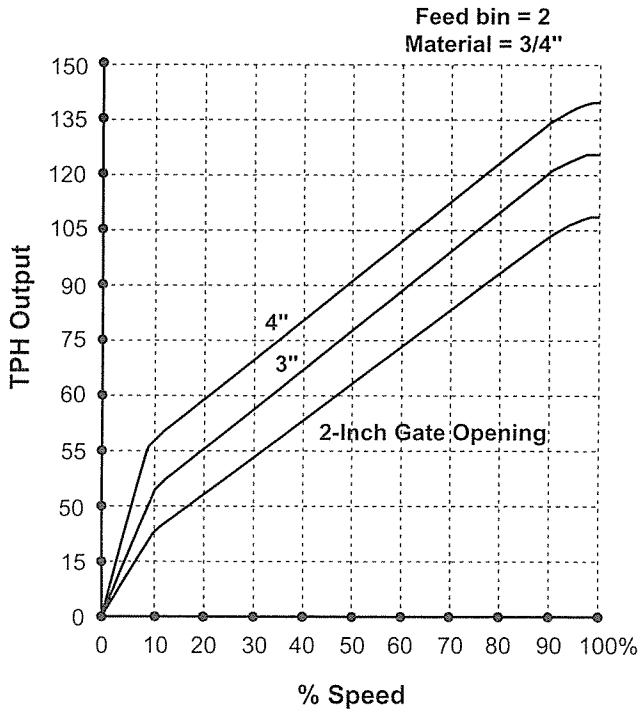


Figure 7.27 Manual Calibration Chart for Cold-Feed Bin

have to be opened, and for very low quantities of material, the gate will have to be closed. The different lines represent the different gate openings.

When drum-mixers first arrived on the hot mix production scene, microprocessor controls for plant operation did not exist. Feeders, or cold feeds, therefore, were typically calibrated with charts similar to that shown in Figure 7.27. The operator, using these types of charts, had to calculate the required speed for each feeder, mix recipe, and desired production rate. Since the 1980s, computerized control (see Figure 7.28) has become the standard for both batch and drum production.

The operator is also given a mix recipe screen, where the desired recipe is entered. A copy of such a screen is shown in Figure 7.29. In running mix, the operator selects the Mix Design Formula (MDF) and the production rate. The processor calculates the required feeder speeds for each feeder based on both the MDF and the desired rate. When the plant is started, the feeders start individually in sequence, which minimizes wasted mix on startup and shutdown. The processor adjusts the feeders to the proper speed setting for the production rate and mix design.

Feeders are calibrated in dry weights because the same amount of aggregate flows from the feeder at a given speed, regardless of the surface moisture. For proper ratio blending, then, it is important to subtract the surface moisture when the calibration process is executed. Figure 7.28 illustrates the means of entering the moisture content during the calibration process.

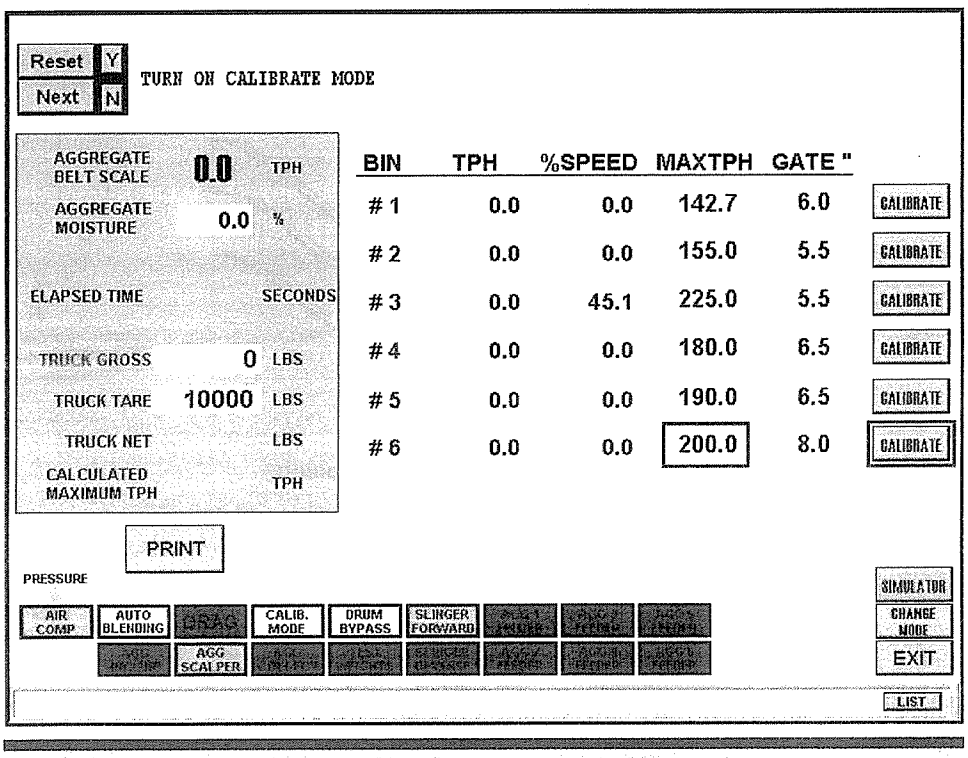


Figure 7.28 Computerized Cold-Feed Calibration Screen

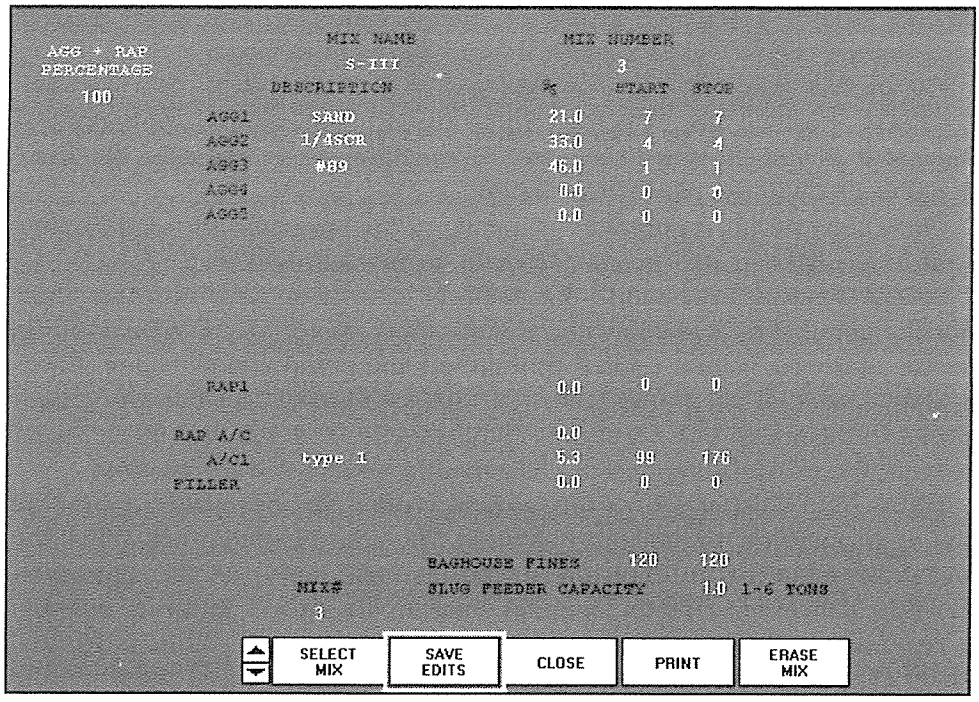


Figure 7.29 Computerized Mix Formula Screen

With a drum-mix plant, the aggregate belt scale is used for blending the asphalt and other ingredients to the combined-aggregate flow. The belt scale is the device that measures the combined-aggregate flow. The presence of a belt scale makes it very easy to calibrate the feeders. The flow of the material from each bin can be easily determined in tons per hour at different flow rates by simply allowing the aggregate material from the cold-feed bin to flow over the belt scale and into a truck. Timed runs can also be used, and the flow rate calculated, but using the belt scale is very simple. Virtually all plants use this technique for calibrating the feeders. The aggregate flow and gradation is controlled volumetrically by using predetermined flow output for different speed ranges and then back-calculating the required speed for each feeder after determining the flow required from each feeder. Executing the feeder calibration process properly is critical for success with a drum-mixer. Feeder calibration should also be checked and re-verified on a regular basis.

7.4.3 Asphalt Content Control in a Drum-Mixer

As indicated above, the belt scale is used to indicate total aggregate flow from the cold feed bins. This is the first step in asphalt content control with a drum-mixer. Once the total aggregate flow is determined, the required

asphalt flow is calculated based on the mix design. The plant control system then regulates the asphalt flow to the required rate.

The asphalt pumping and metering system on a drum-mix plant is a continuously operating unit. It pumps liquid asphalt continuously and measures the result of that flow with a meter. It can increase or decrease the flow dynamically with the plant's automation.

There are two key equipment elements to this process—the asphalt flow control mechanism and the asphalt meter or measuring device.

Several types of flow control mechanisms have been used over the years, but all manufacturers now favor a variable-speed motor on the asphalt pump for this purpose. If more asphalt is required, the pump is accelerated. If less asphalt is needed, the pump is slowed. Most producers have converted their plants to this technique because it eliminates the maintenance on the wear parts of the previous systems and is generally considered trouble-free.

Figure 7.30 illustrates how this process works. The belt scale reports the wet aggregate flow to the control system. The control system will adjust the wet scale value for the moisture in the material, based on the aggregate-moisture data entered by the operator. Entering the proper moisture quantity is important and is typically done by entering the moistures of the individual materials into a "moisture screen." The control system then calculates the combined moisture based on the mix recipe and the individual material moistures, and removes this combined moisture percentage from the wet weight to arrive at a dry weight. The plant control assumes the dryer will dry the material adequately and therefore uses this calculated dry weight signal for proportioning the liquid-asphalt binder. Based on the required asphalt percentage in the mix, which comes from the mix formula, the control system calculates the required asphalt flow. It then sends a signal to the flow-control mechanism, which more often than not is a variable-speed motor, and then reads the result of its action by reading the asphalt flow meter. If

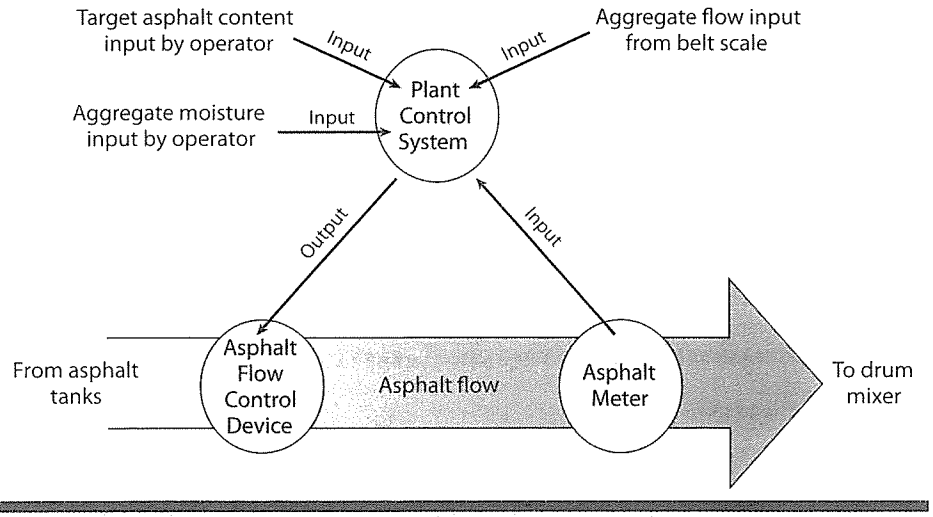


Figure 7.30 Drum-Mix Asphalt Flow Control

the asphalt flow matches the requirement, no change is made. If the asphalt flow is either too high or too low, the control system will correct the asphalt flow by adjusting the asphalt flow-control mechanism until it reaches the right target.

If the aggregate flow is increased, a new target asphalt flow rate is calculated and the control system will adjust the asphalt flow. If the aggregate flow decreases, the opposite happens. This entire process is continuous.

Changeable delay times are built into the flow-control circuits. When the plant is set up, tests are done to determine the length of the delay time between the belt scale reading and the aggregate material reaching the asphalt injection point for that plant. Therefore, changes in the aggregate scale do not cause an instantaneous change in the asphalt flow. The plant control system has delay times set up not only for the asphalt flow but also for each cold-feed bin and for every other ingredient. These flow delays are very important for a continuous process. When changes to production are made, mix integrity must be maintained, so the delay times are carefully calibrated to ensure that production-rate changes and mix changes can be executed without shutting down the process. An alarm notifies the operator when a mix change is complete and material can be directed to another storage silo.

Ingredients other than asphalt binder, including other liquid additives and bulk additives, are controlled in the same fashion as the liquid asphalt binder. They all take their key from the aggregate belt scale signal. When mixes containing RAP are being produced, the control system also reads the belt scale on the RAP conveyor and adjusts the virgin-asphalt binder to compensate for the asphalt that is included in the RAP.

The belt scale reading is very critical in producing quality HMA in the drum-mixer process. Additional attention needs to be paid to the area of the main belt where the belt scale is located. Before startup, the area should be inspected and cleaned to make sure the framework is free of dust and dirt buildup that may affect the readings. Wind screens mounted around the belt scale will prevent wind loading that can artificially add or subtract weight from the readings. A belt tensioner or deadman (weighted idler roller) to take up slack as the belt stretches helps to keep constant tension on the belt as it passes across the belt scale and helps improves the accuracy of the readings.

7.4.4 Mixing/Coating with a Drum/ Continuous Plant

Located in the rear of the drum-mixer or in the outside mixer are mixing flights, or tips, for mixing the aggregate materials with the liquid asphalt binder.

Figure 7.31 shows mixing flights in a one-piece drum-mixer plant. Figure 7.32 shows the outside mixing tips in a double-barrel unitized

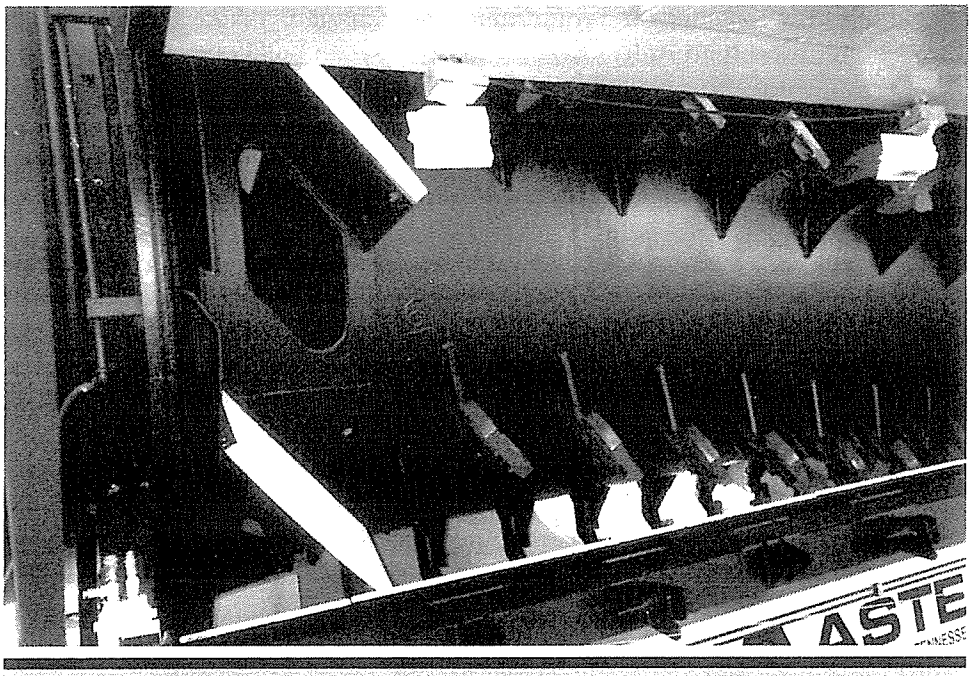


Figure 7.31 Mixing Flights/Drum-Mixer Plant

dryer-mixer plant. Mixing flights in plants with separate mixing devices are configured similarly and meet the same objective—that of mixing the liquid asphalt with the aggregate materials to produce a homogenous mix. Retention time in the mixing area of a drum-mixer is typically 1–3 minutes. The test described in Section 7.3.3, "Mixing/Coating with a Batch Plant," for measuring uncoated particles, can be used on material produced in a drum-mix plant.

If mixing flights are not cleaned periodically, very hard pieces of asphalt and fines, which build up on the tips and flights, can break loose and find their way into the mix. Mixing effectiveness can also be affected. If mixing flights are allowed to build up with material between the flights to the point where flights are barely visible, segregation can occur in the mixing process. Also larger aggregate particles can make it through the mixing area without being adequately tumbled and coated.

7.4.5 Recycled Mix Production with a Drum-Mixer

With a drum-mixer plant, Reclaimed Asphalt Pavement (RAP) is introduced directly into the drum-mixer dryer after the virgin aggregates are introduced. To the casual observer, RAP is being introduced into the drum similar to the virgin aggregates, and the process does not seem any different. What is actually occurring inside the drum-mixer is that the virgin aggregate materials are being super-heated, and these super-heated aggregates are

then being used to dry and heat the RAP to produce the new recycled hot mix at the required final temperature. The burner control system is used to regulate the temperature of the final combined product, just as it is used to control the final temperature of mix produced from only new aggregates. The steam released from the RAP is evacuated from the drum-mixer with the steam from the virgin aggregates.

One or more cold-feed bins are dedicated for the RAP products, but a separate belt scale is used to measure the quantity of RAP introduced into the system. The control system uses this scale figure to control the ratio of RAP and virgin-aggregate materials, along with the quantity of new binder based on the quantity of usable binder in the RAP.

More information on producing recycled mixes in a drum-mixer plant is available from the National Asphalt Pavement Association in Information Series 123, "Processing and Handling Reclaimed Asphalt Pavements."

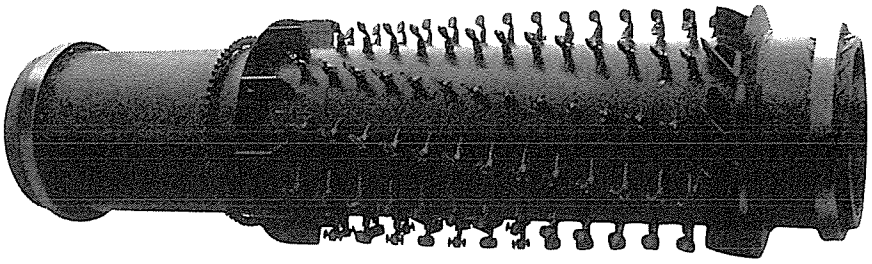


Figure 7.32 Mixing Tips/Double-Barrel Unitized Dryer/Mixer Plant

7.5 Mix Deficiencies and Problem Solving

Successful mix production requires a diligent approach to both the calibration process and daily operational best practices. For successful placement, a mix must have accurate and consistent aggregate gradation, asphalt content, and temperature. The success of the in-place product starts with the mix-production process.

The entire production, transport, and placement process should run smoothly and consistently. Starts and stops increase the likelihood of mix inconsistencies at the plant, whether a drum-mix or batch plant is used. Starts and stops in placement also affect the quality of the placed product. Every activity—from scheduling the production rate of the plant, to scheduling the lay-down and compaction rate of the placement crew, to selecting the number of trucks for the project, to the cycling of those trucks—should be as organized as possible. Success will directly correlate to not only the proficiency in mix production and the proficiency of the placement crew but also to the smooth coordination of each stage of the process.

Even with the best intentions, things can turn out wrong. Sometimes equipment fails. Often the failures are operational. When test results and mix results are not as anticipated, it is useful to have a troubleshooting or problem-solving guide to help find the cause of the problem so it can be corrected.

The following two sections provide problem-solving guides, one for drum-mix plant production and one for batch-plant production. They are organized by the types of tests typically performed on the mix—gradation analysis, asphalt content analysis, voids calculation, etc. The two sections do not cover every possible cause of every possible mix deficiency, but they do cover most common problems found in the field.

7.5.1 Drum-Mix Production Problem-Solving Guide

The following guide will prove useful in solving problems with mixes produced by a drum-mix plant. Common mix deficiencies and common causes are discussed.

Gradation Shift/Aggregates

Check for the following:

- Cold-feed bin flow restriction/change
- Cold-feed bin calibration shift
- Gradation change in feed aggregates
- Improper loader management technique on stockpiled material

Gradation Shift/ Coarser on –200 Mesh (75 Micron) Sieve

Check for the following:

- Drastic change in production rate (low to high takes time for baghouse fines to catch up to cold-feed change)
- Overdrafting dryer (less fines in mix from damper being more open relative to previous or typical production rate, together with baghouse fines being wasted or partial return)
- Baghouse dust return or mineral filler additive malfunction (causing reduced baghouse dust return or mineral filler additive)
- Gradation change in feed aggregates (see above)

Gradation Shift/ Finer on –200 Mesh (75 Micron) Sieve

Check for the following:

- Drastic change in production rate (high to low takes time for baghouse fines to catch up to cold-feed change)
- Underdrafting dryer (more fines in mix from damper being less open relative to previous or typical production rate, together with baghouse fines being wasted or partial return)
- Baghouse dust return or mineral-filler additive malfunction (causing increased baghouse dust return or mineral additive)
- Gradation change in feed aggregates (see above)

Low Air Voids

Check for the following:

- Dust-return or dust-additive malfunction (see above)
- Production-rate change (high to low with immediate sampling)
- Lack of draft in dryer—retained fines in dryer/primary baghouse
- Gradation change in feed aggregates (see above)
- AC content high (see below)
- Specific gravity change in aggregates

High Voids

Check for the following:

- Dust-return or dust-additive malfunction (see above)
- Production-rate change (low to high with immediate sampling)
- Overdrafting dryer—more fines to baghouse versus in dryer/primary (and baghouse fines are being wasted or used partially)
- Gradation change feed aggregates (see above)
- AC content low (see below)
- Specific gravity change in aggregates

AC Content High

Check for the following:

- Calibration of belt scale (reading high)
- Binding belt scale (reading high)
- Wind effect on belt scale (reading high)
- Calibration of AC meter (reading low)

- Improper moisture setting (too low)
- Wrong AC weight/gallon entry with gallon meter (too low)
- Inadequate drying of aggregate (drying efficiency low)
- Mixing flights inadequate (mix not homogenous)

AC Content Low

Check for the following:

- Calibration of belt scale (reading low)
- Binding belt scale (reading low)
- Wind effect on belt scale (reading low)
- Calibration of AC meter (reading low)
- Improper moisture setting (too high)
- Wrong AC weight/gallon entry with gallon meter (too high)
- Mixing flights inadequate (mix not homogenous)

AC Content Varies

Check for the following:

- Malfunction in flow-control mechanism (difficulty in regulating/mechanical issue typically)
- Improper moisture settings (not changing entry with moisture changes)
- Inadequate drying of aggregate (drying efficiency varies and ignition oven AC test method removes water in addition to AC)
- Inaccurate AC weight/gallon entry with gallon meter (not changing entry with AC changes)
- Mixing flights inadequate (not tall enough, caked, need different design)

Inadequate Coating

Check for the following:

- Asphalt content low (not enough asphalt binder)
- AC injection point too close to discharge (inadequate mixing time)
- Mixing flights inadequate (not tall enough, caked, need different design)
- Temperature of liquid-asphalt binder too low
- Insufficient moisture in aggregate

Temperature Fluctuations

Check for the following:

- Worn dryer flights (incapable of drying aggregates sufficiently at all production rates)
- Moisture fluctuations in feed aggregates (burner/dryer cannot rapidly adjust to rapidly adjusting moisture variations)
- Excessive storage time (mix cooling off from extended storage)
- Mix thermocouple malfunctioning/not reading correctly
- Mix thermocouple covered with material and/or not properly positioned in material stream

7.5.2 Batch-Plant Production Problem-Solving Guide

The following guide is useful for solving batch-plant mix-deficiency issues. Common mix deficiencies and causes are discussed.

Gradation Shift/Aggregates

Check for the following:

- Hole in screen
- Hole in screen exit chute
- Hole in walls between hot bins
- Cold-feed ratio/blend change (affecting hot-bin gradations)
- Cold-feed bin flow restriction (affecting hot-bin gradations)
- Cold-feed bin calibration shift (affecting hot-bin gradations)
- Gradation change in feed aggregates (affecting hot-bin gradations)
- Improper loader management technique on stockpiled material (affecting hot-bin gradations)

Gradation Shift/Coarser on –200 Mesh (75 Micron) Sieve

Check for the following:

- Hot-bin stratification from vibration/idling
- Drastic change in production rate (low to high takes time for baghouse fines to catch up to cold-feed change)
- Overdrafting dryer (less fines in mix from damper being more open relative to previous or typical production rate, together with baghouse fines being wasted or partial return)
- Baghouse dust-return or mineral-filler additive malfunction (causing reduced baghouse dust-return or mineral-filler additive)
- Gradation change in feed aggregate(s) (see above)
- Cold-feed ratio/blend changed, affecting hot-bin gradations (see above)

Gradation Shift/Finer on –200 Mesh (75 Micron) Sieve

Check for the following:

- Dust slough (slide) in fines bin
- Hot-bin stratification from vibration/idling
- Drastic change in production rate (high to low takes time for baghouse fines to catch up to cold-feed change)
- Underdrafting dryer (more fines in mix from damper being less open relative to previous or typical production rate, together with baghouse fines being wasted or partial return)
- Baghouse dust-return or mineral-filler additive malfunction (causing increased baghouse dust-return or mineral-filler additive)
- Gradation change in feed aggregate(s) (see above)
- Cold-feed ratio/blend changed, affecting hot-bin gradations (see above)

Low Voids

Check for the following:

- Dust slough (slide) in fines bin
- Hot-bin stratification from vibration/idling
- Dust-return or dust-additive malfunction (see above)
- Production-rate change (high to low with immediate sampling)
- Lack of draft in dryer—retained fines in dryer/primary versus to baghouse (and baghouse fines are being wasted or used partially)
- Gradation change in feed aggregates (see above)
- Cold-feed ratio/blend changed, affecting hot-bin gradations (see above)
- AC content high (see below)
- Specific gravity change in aggregates

High Voids

Check for the following:

- Hot-bin stratification from vibration/idling
- Dust-return or dust-additive malfunction
- Production-rate change (low to high with immediate sampling)
- Overdrafting dryer—more fines to baghouse versus in dryer/primary (and baghouse fines are being wasted or used partially)
- Gradation change in feed aggregates (see above)
- Cold-feed ratio/blend changed, affecting hot-bin gradations (see above)
- AC content low (see below)
- Specific gravity change in aggregates

AC Content High

Check for the following:

- Malfunction in weigh bucket
- Malfunctioning AC injection/coating and sample in rich side of batch
- Missing pugmill tips, inadequate mixing and sample in rich side of batch
- Weigh bucket out of calibration
- Automation error/consistently miss target (free fall compensation?)

AC Content Low

Check for the following:

- Malfunction in weigh bucket
- Malfunctioning AC injection/coating and sample in lean side of batch
- Missing pugmill tips, inadequate mixing and sample in lean side of batch
- Weigh bucket out of calibration
- Automation error/consistently miss target (free fall compensation?)

AC Content Varies

Check for the following:

- Malfunction in weigh bucket
- Malfunctioning AC injection/coating and sampling detects variation in homogeneity

- Missing pugmill tips; inadequate mixing
- Automation error/inconsistency

Inadequate Coating

Check for the following:

- Too short wet-mix cycle time
- Missing pugmill tips
- Tip-to-liner clearance (too great)
- Inadequate drying of aggregate
- Dust slough/dry batch
- Temperature of liquid-asphalt binder too low

Temperature Fluctuations

Check for the following:

- Worn dryer flights (incapable of drying aggregates sufficiently at all production rates)
- Moisture fluctuations in feed aggregates (burner/dryer cannot rapidly adjust to rapidly adjusting moisture variations)
- Excessive storage time (mix cooling off from extended storage)
- Mix thermocouple malfunctioning/not reading correctly
- Mix thermocouple covered with material and/or not properly positioned in material stream

Resources

1. MS-5, *Introduction to Asphalt*. The Asphalt Institute, Lexington, KY.
2. MS-22, *HMA Construction*. The Asphalt Institute, Lexington, KY.
3. *Hot Mix Asphalt Paving Handbook 2000*. National Asphalt Pavement Association, Lanham, MD.
4. *Processing and Handling Reclaimed Asphalt Pavements (RAP)*" Information Series 123. National Asphalt Pavement Association, Lanham, MD.
5. Roberts, Freddy L., Kandhal, Prithvi S., Brown, E. Ray, Lee, Dah-Yin, Kennedy, Thomas. *Hot Mix Asphalt Materials, Mixture Design and Construction*. NAPA Education Foundation, Lanham, MD.

CHAPTER 8

HOT MIX ASPHALT PAVING

This chapter describes paving fundamentals, beginning with surface preparation and including delivery of the hot mix asphalt (HMA) to the paving site and the operation of the paving machine. It also covers a variety of related paving operations topics.

Section 8.1 Pre-paving Conference

Section 8.2 Preparing the Surface for Paving
- 8.2.1 Surface Preparation and Protection
- 8.2.2 Preparation of Existing Pavement for Overlay
- 8.2.3 Horizontal and Vertical Grade Control
- 8.2.4 Prime Coats and Tack Coats

Section 8.3 Delivery of HMA
- 8.3.1 Types of Haul Trucks
- 8.3.2 Delivery of HMA to the Paver
- 8.3.3 Control of Material Surge in Front of Paver
- 8.3.4 Load Tickets
- 8.3.5 Visual Inspection of the HMA

Section 8.4 Paving Equipment
- 8.4.1 Asphalt Paving Machine–Basics
- 8.4.2 Types of Tractors
- 8.4.3 Screeds
- 8.4.4 Automatic Screed Controls (Types and Operational Differences)
- 8.4.5 Leveling Effect of the Paver and the Free-Floating Screed
- 8.4.6 Fumes Control
- 8.4.7 Observation of the HMA Pavement Behind the Screed

Section 8.5 Joint Construction
8.5.1 Transverse or Construction Joints
8.5.2 Paving Start-up
8.5.3 Longitudinal Joints
8.5.4 Notched-Wedge Joints

Section 8.6 Related Paving Operations
8.6.1 Hand-Spreading Operations
8.6.2 Supplemental Operations and Appurtenances

Section 8.7 Foreman's/Inspector's Check Sheet

8.1 Pre-paving Conference

The key to a successful paving project is good communication between the paving crew, plant personnel, and the agency or owner. This communication generally begins with the pre-paving conference. At this conference, the contractor's personnel should explain their project plans and establish a meaningful schedule to meet the agency's or owner's expectations.

The details of the project should be covered so that all the affected parties understand the "who, what, why, when, and how." The pre-paving conference can be as simple as the contractor's representative meeting with the owner on the project site to explain his plan to pave the project—or as complicated as an organized meeting with all interested parties (including the owner, the designer, the inspectors, personnel from the testing firms, the contractor's plant and paving personnel, utility representatives, etc.). No matter what form this pre-paving meeting takes, it is very critical and will save time, money, and confusion provided it is held well before HMA placement begins.

The discussion of the project details allows everyone involved with the project to identify concerns they may have with the proposed schedule and the possible impacts on pavement users (in terms of lane closures), traffic patterns, on businesses adjacent to the project, and local or regional events such as major sporting events.

The most important item to be discussed at the pre-paving conference is safety—of the construction personnel as well as of the traveling public. Clear responsibility for the maintenance of traffic-control devices, such as pavement marking and flagging, should be delineated. Attention to project safety is a moral, legal, and financial responsibility of the contracting owner or agency, the contractor, all subcontractors, and suppliers.

Testing and sampling methods and frequencies should be reviewed at the pre-paving conference to ensure that everyone involved in the project understands the following:

- The purpose of the testing
- The frequency of the testing
- Who will conduct the tests
- How and to whom the test results will be reported

8.2 Preparing the Surface for Paving

8.2.1 Surface Preparation and Protection

A properly prepared foundation, which includes adequate surface and subsurface drainage and proper compaction during construction, will help to ensure a long life for a pavement structure. Proper drainage is extremely important. The necessity for lowering the water table below the depth of frost penetration or the bottom of the improved subgrade by lateral drainage or subsurface drainage is generally accepted by paving technologists as an essential element of good construction.

Proper compaction of each layer of embankment, subgrade, improved subgrade, and base course is equally important. Compaction greatly increases the supporting power of the subgrade. When the subgrade is not sufficiently compacted, additional consolidation may occur under traffic, resulting in settlement and possible failure.

The following roadway surfaces are generally considered unpaved surfaces:

- Subgrade
- Improved subgrade
- Untreated base
- Nonsurfaced granular roadway

Subgrade

Compaction tests are made in the laboratory on materials to be used in construction to determine the maximum practical density that can be obtained. Laboratory densities are determined using AASHTO T 180. Recommended subgrade compaction criterion are as follows:

- **Cohesive soils**—minimum 95 percent density in top 12 inches (300 millimeters) and minimum 90 percent for all fill areas below the top 12 inches (300 millimeters)
- **Cohesionless soils**—minimum 100 percent density in top 12 inches (300 millimeters) and minimum 95 percent for all fill areas below top 12 inches (300 millimeters)

A specifying engineer should verify these requirements with the appropriate local, state, or in some cases, federal agency where the project is located.

The selected moisture content for compacting cohesive soils should provide the highest remolded strength consistent with expansion considerations. Specifications usually require the moisture content to be within ±2 percent of the optimum moisture content. Generally, fine-graded nonexpansive soils should be compacted to 1 or 2 percent below the optimum moisture content for best results. Fine-graded expansive soils

should be compacted at 1 to 3 percent above the optimum moisture content to reduce the potential for swelling. The compaction criteria should apply to both cut and fill sections. In some instances, this requires scarifying, adding moisture, replacement, and recompaction to produce uniformity in the subgrade and to avoid differential settlement.

Although compaction increases the stability of most soils, some soils decrease in stability when scarified, worked, and rolled. Some soils also shrink excessively during dry periods and expand excessively when allowed to absorb moisture. They may also lose strength when reworked—for example, silts that can become "quick" when remolded. When these conditions exist, special treatment is required under the supervision of a qualified geotechnical engineer.

Surface irregularities in an HMA pavement structure can result from traffic densification of the underlying courses, usually in the wheel paths. Such irregularities indicate that one or more courses were not adequately compacted during construction or that soft, unstable areas exist in the structure. To detect these deficiencies, proof rolling of the subgrade is sometimes performed during construction. Proof rolling is the application of heavy, pneumatic-tire rollers or large loaded dump trucks over the subgrade, usually following initial compaction by conventional means. Proof rolling accomplishes two purposes: it locates unstable areas and achieves additional compaction.

Prepared Subgrade

A prepared subgrade is one that has been worked and compacted. This may be the foundation soil or a layer of stabilized soil, select soil, or otherwise improved subgrade material. The riding quality of the pavement surface depends largely on proper construction and preparation of the foundation material. The roadway should be shaped and rolled so that the paving equipment has no difficulty in placing material at a uniform thickness and smooth grade. Weather conditions should be suitable, and the roadway surface should be firm, dust-free, dry, or just slightly damp when paving operations are started.

Untreated Base

Untreated granular bases should be properly shaped and compacted immediately before paving. This procedure will restore any proposed line or grade that may have been damaged during construction. When asphalt pavement courses are to be placed on an untreated aggregate base, loose granular particles and dust should be lightly swept from the surface. Care should be taken, however, not to dislodge or otherwise disturb the bond of the aggregate in the surface of the base. When the loose particles have been properly removed, only the tops of the aggregate in the surface will be exposed. After the surface has been cleaned, it is ready to be primed with asphalt. For a discussion about prime coats, see Section 8.2.4, "Prime Coats."

Nonsurfaced Granular Roadway

If an HMA pavement is placed on an existing granular roadway, the surface should be shaped similar to untreated granular bases. The granular roadway differs primarily in that it has been used as a roadway by traffic and has typically been in use for a considerable time. It is advisable, when the granular roadway is rough and uneven, to bring the surface to the proper line and grade by scarifying the top few inches, blending in more aggregate material as necessary, compacting, and then priming with asphalt. If the granular roadway has had a previous asphalt treatment and is rough and uneven, but structurally sound, it is advisable to place a leveling course of HMA ahead of the first layer of the surface course. Otherwise, the surface may have to be recycled by scarifying the surface and mixing it with a portion of the existing base. The engineer should inspect the surface well in advance of placing the surface courses and determine what grade corrections need to be made. As with untreated granular bases, the surface should be cleaned and swept before initiating paving operations, including any leveling courses that may be needed.

8.2.2 Preparation of Existing Pavement for Overlay

The amount of preparation needed for an existing HMA pavement before overlay will depend on the condition and the type of distress in the existing pavement. Portions of the pavement that have "failed" (fatigue/alligator cracking) should be removed and replaced with full-depth HMA. Potholes need to be properly repaired, cracks need to be sealed or filled, and rutting areas must be removed by cold milling. If these distresses cover more than 40 percent of the pavement area, the use of cold in situ recycling or full-depth reclamation should be considered. The procedures for proper pre-overlay repairs are described in Chapter 12, "Maintenance of Asphalt Pavements," and the procedures for cold milling, recycling, and full-depth reclamation are described in Chapter 13, "Rehabilitation Methods for Asphalt Pavements." The repair procedures for Portland cement concrete (PCC) pavements are described in Chapter 14, "Rehabilitation of Concrete Pavements with HMA."

8.2.3 Horizontal and Vertical Grade Control

Construction of a high quality smooth HMA pavement requires that the subgrade and base course be built to the correct horizontal and vertical grades. Before placement of the first lift of the HMA pavement on the base course, elevations should be checked to ensure that the correct grades have been achieved. Grades for aggregate bases should generally be within 0.05–0.10 foot (15–30 millimeters) of the plan elevations. Additionally, the surface

should not have any deviations greater than ¼–½ inch (6–12.5 millimeters) measured from the bottom of a 16-foot (4.9 meters) straightedge anywhere along its length.

8.2.4 Prime Coats and Tack Coats

Prime Coats

A prime coat is an application of a penetrating liquid asphalt (cutback, specially designed emulsion, or other suitable material) applied with a distributor (see Figure 8.1) to a compacted aggregate base course. The function of a prime coat is to provide a protective layer to prevent the surface of the aggregate base course from raveling under construction traffic. It also provides a waterproof layer, preventing it from absorbing or losing excess moisture before paving. A prime coat helps bond the top layer of the granular base to the first layer of HMA pavement. Although this bonding helps prevent slippage between the two layers, it is very weak and should not be considered when overall strength of the pavement is analyzed. The prime coat is applied to an absorbent surface, such as a granular base or prepared subgrade, with an asphalt distributor truck. A cutback asphalt (such as MC-30, MC-70, or a specially designed emulsified asphalt) is typically applied at a rate of 0.20–0.50 gallons per square yard (0.9–2.3 liter per square meter) on the prepared surface. See Chapter 12, "Asphalt

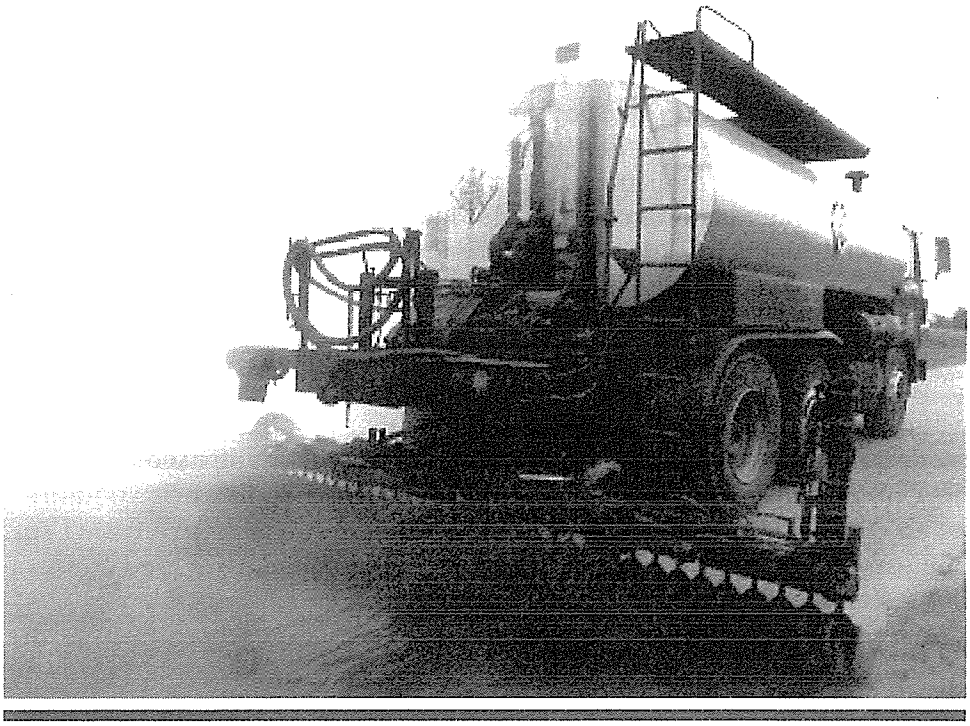


Figure 8.1 Distributor Applying a Prime Coat

in Pavement Maintenance," for a discussion of the proper operation of the distributor. The asphalt grade and application rate will depend on the surface condition and permeability of the base layer being treated. The asphalt penetrates the surface and should be entirely absorbed by the base layer. If the asphalt is not absorbed within 24 hours after application, a light coating of clean dry sand should be spread over the surface to blot the excess and the application rate adjusted to prevent applying excessive asphalt. The prime coat should be fully set and cured before placing the HMA. A prime coat should not be applied when the surface is wet, the air temperature is below 50°F (10°C), or the air temperature over the preceding 24 hours has been 40°F (4.5°C) or less.

In many areas, the use of cutback asphalt is not permitted because of the release of volatiles into the atmosphere. Local air quality requirements should be checked to determine if cutbacks can be used. As a substitute, asphalt suppliers have developed special penetrating emulsion materials for use as prime coats, which are spray applied in the same manner as cutback asphalts.

Tack Coats

Tack coats are a very light application of liquid asphalt applied with a distributor to an existing nonporous surface. The purpose of a tack coat is to promote bonding between the underlying surface and the new HMA layer. Normally, an asphalt emulsion or hot asphalt cement is used for a tack coat, but some agencies allow the use of other asphalt products. The tack coat should be placed on a clean, dry pavement, free of any materials that might prevent bonding. The application rate will vary depending on the type and condition of the surface being overlaid. The goal is to place a thin, uniform coating of asphalt covering 90–100 percent of the pavement surface. The application rate for a tack coat is normally 0.05–0.15 gallons per square yard (0.25–0.70 liter per square meter) of diluted slow-setting cationic or anionic emulsion (SS-1, SS-1h, CSS-1, or CSS-1h). A diluted emulsion is an emulsion that has been diluted using one part water to one part asphalt emulsion. If a given emulsion contains 65 percent residual asphalt in its original state, then after a 50-50 dilution, the product applied to the roadway would contain a 32.5 percent residual asphalt. Spread rate and uniformity are hard to control when using an asphalt distributor set at a very low application rate. By diluting the tack coat, the emulsion can be applied at a much higher spread rate while providing a uniformly thin coating of residual asphalt. A low application rate is usually used between new or un-trafficked pavement layers. An intermediate rate is required when resurfacing an existing, relatively smooth, asphalt pavement. The highest rate is used on old oxidized, cracked, pocked, or milled HMA or PCC pavement surfaces.

Distributor operations should be monitored very closely to prevent the accidental application of too much asphalt, which commonly occurs in those areas where the distributor starts and stops. If this happens, the distributor should be prevented from applying tack until the problem with

the distributor or its operations are corrected. The excess material should be squeeged over a larger area to thin out the excess to a more acceptable application rate.

The asphalt emulsion should break and cure before placement of asphalt concrete. Shortly after placement of the tack coat, the emulsion will break. The color will change from brown to black, and the water will begin to evaporate. Once the water has evaporated, the mechanical properties of the residual asphalt will return and result in a strong adhesive bond. The rate of this breaking-and-setting process will depend on the environmental conditions. Generally, the emulsions used for tack coats will break and be fully cured shortly after placement. During this process, equipment (rollers, delivery trucks, etc.) should be kept off the surface. Tack coats leaving surplus asphalt on the surface can form a slip plane. Excessive asphalt may also work its way up into the overlying course, which may eventually flush asphalt to the surface. A light tack coat does no harm and it bonds the surfaces, a necessity for good construction. Work should be planned so that no more than the necessary tack coat for the paving shift is placed on the surface. All traffic not essential to the work should be kept off the tack coat.

Vertical faces of existing pavements such as curbs and gutters should also be sprayed or painted with a uniform coating of approved tack material. This work should be done such that exposed curb or gutter surfaces are not stained. The use of low-penetration tack material such as an SS-1h or CSS-1h will also reduce the amount of tracking. Tracking is the unwanted coating or staining of adjacent pavement surfaces, both on and off the project, by trucks or equipment tires.

Asphalt Distributor

The asphalt distributor and the operation of the distributor are covered in detail in Chapter 12, "Maintenance of Asphalt Pavements."

8.3 Delivery of HMA

A continuous flow of HMA material from the plant to the paver ensures a consistently high-quality HMA pavement. The tonnage to be produced, the size and types of trucks, the time to make one trip from plant to paver, how much HMA to be placed and compacted—all are details that must be considered. These parameters dictate the necessary number of trucks. The time between each delivery must be maintained at a consistent level, and the trucks must not be allowed to bunch up at the paver or at the plant. Trucks may be delayed in traffic (construction traffic control can be a major cause of backups that delay the material delivery trucks). If a break in the delivery of material occurs, the paver should be stopped using appropriate techniques such as the rapid stop/start method (the paver is stopped and started without an extended acceleration/de-acceleration period). Paving should not resume until a sufficient number of trucks are on the job site to ensure nearly continuous paving. Starting and stopping the paver for each truck as it arrives on the project will reduce the smoothness and quality of the finished pavement.

8.3.1 Types of Haul Trucks

Three types of haul vehicles are used to transport HMA from the plant to the paver: end-dump, bottom- or belly-dump, and live-bottom. Proper trucking procedures are necessary for maintaining a continuous flow of material to the paver hopper.

End-Dump Trucks

The end-dump truck (see Figure 8.2) delivers mix directly to the paver hopper and may have three to six axles. It can have a capacity of 12 to 20 tons (11 to 18 tonnes). More axles mean more capacity. The advantage of the end-dump truck is its short wheelbase, which makes it easy to maneuver. The end-dump truck can also be a semitrailer truck (see Figure 8.3), which is essentially a large dump truck, with a capacity of 20 to 25 tons (18 to 23 tonnes). The disadvantage with this type of truck is that improved operator skill is needed and its large dump bed can be the source of problems with overhead obstructions (wires, bridges, and trees). Because of this truck's size, special care must be exercised when loading the truck because of the increased potential for aggregate segregation.

Bottom- or Belly-Dump Trucks

Bottom-dump trailers (see Figure 8.4) used in conjunction with a windrow pickup machine require proper cleaning and maintenance. The bottom dump doors must be cleaned and sprayed with a proper release agent